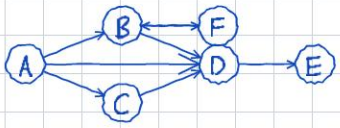
Depth-First Search (DFS)

DFS explores as far as possible along each branch before backtracking. It uses a stack (implicit via recursion or explicit) to track the frontier.



Recursive Algorithm

```
DFS(G):
  time ← 0
  for each vertex u in G:
    color[u] ← WHITE
  for each vertex u in G:
    if color[u] == WHITE:
      DFS-VISIT(u)

DFS-VISIT(u):
  time ← time + 1
  d[u] ← time              // discovery
  color[u] ← GRAY
  for each v in Adj[u]:
    if color[v] == WHITE:
      parent[v] ← u
      DFS-VISIT(v)
  color[u] ← BLACK
  time ← time + 1
  f[u] ← time              // finish
```

Discovery / Finish Times (example above, starting from A)

Vertex	d	f
A	1	12
B	2	9
C	10	11
D	3	6
E	4	5
F	7	8

Edge classification

For each edge $(u \to v)$, classify by color[v] when first explored:

- Tree edge – v is WHITE (discovered via this edge). Forms the DFS forest.
- Back edge – v is GRAY (ancestor still on stack). Indicates a cycle.
- Forward edge – v is BLACK and v is a descendant of u.
- Cross edge – v is BLACK and v is NOT a descendant of u.

In our example:

- Tree: A→B, B→D, D→E, B→F, A→C
- Back: F→B (B is gray, on the recursion stack)
- Forward: A→D (D is black, descendant of A)
- Cross: C→D (D is black, not a descendant of C)

Key rule: $u.d < v.d < v.f < u.f$ iff v is a descendant of u in the DFS tree.

Applications

- Cycle detection: a graph has a cycle iff DFS finds a back edge.
- Topological sort: process vertices by decreasing finish time (reverse of finish order). Valid only for DAGs.
- Strongly connected components (SCCs): run DFS, transpose graph, process vertices in decreasing finish order (Kosaraju's or Tarjan's algorithm).
- Path finding, maze solving, connected components in undirected graphs.

Time complexity

Each vertex is discovered and finished once → $O(V)$.
Each edge is examined once (directed) or twice (undirected) → $O(E)$.

Total: $\boxed{O(V + E)}$

This is optimal - we must examine every vertex and edge at least once.

DFS goes deep first - use recursion or explicit stack